

Atmospheric Corrosion Tests of Metals in SO₂-Polluted Cold Atmosphere in Northern Norway and along Its Border with Russia

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Abstract—The effect of SO₂ on the corrosion of metals was independently studied in the town of Sulitelma in the northern Norway and along the Russia–Norway boundary in the vicinity of the integrated nickel-production works. Both the investigations confirmed the strong corrosive effect of SO₂ on all the metals tested. The corrosive action of chlorides was also been noted at some sites around the town Nikel'. The effect of temperature on the corrosion was also shown in the investigations and in the joint analysis of results. The temperature limit of the corrosion development in the test sites was in a range of –2 to –4°C. The model based on the test data is capable of forecasting the corrosion of steel for a month, given the monthly values of temperature, relative humidity of air, and the atmosphere's pollution with sulfur dioxide. By using the developed dose–response functions, the atmospheric corrosivity is assessed, and maps of the yearly corrosion of carbon steel and zinc are drawn for the area along the Russia–Norway border of measuring 110 × 100 km at a scale of 5 × 5 km, including tolerable corrosion rates for two corrosion scenarios. The corresponding tolerable levels of sulfur dioxide are determined.

INTRODUCTION

Although corrosion rates in cold climate, are in general relatively low, the cultural heritage of Norway, Russia, Canada, and Antarctica (including Novaya Zemlya, Franz Josef Land, Spitsbergen, Svalbard, Nunavut, Greenland, Alaska, and the peninsula and islands of Antarctica) suffers corrosion damage [1, 2]. At the same time, corrosion tests in cold regions are rare and far from being insufficient. The interest in the corrosion in cold climates is usually focused on its stimulation by chlorides, because the main activity in the polar regions is associated with sea. The Russian program of tests in the Far East may be referred to as one of a few programs realized in the continental regions [3, 4]. The Australian program of researches in Antarctica [5, 6], as well as tests carried out at the northern stations of Norway, Sweden, Canada, and at the Antarctic stations of the MIKAT project [7–10], were performed in the framework of the national and Scandinavian programs. These researches were also envisaged by the ISOKORRAG Technical Committee 156 on the development of international standards in the corrosion field and the MIKAT project on the evaluation of the corrosivity of atmosphere in the Spanish-speaking countries of America, in Spain, and Portugal. It should be noted that all these tests were carried out mainly at the sites with low or very low degrees of air pollution.

By contrast, in the northern areas of Norway and Russia, the mining industry and metallurgy have long

since been developed, and traditionally exerted tremendous negative effects both on the environment (vegetation) and structural materials. The severest environmental consequences of industry in Norway were recorded in the vicinity of the town Rogos, where copper was smelted from 1646 until in 1977, the all-wooden town had been registered in the UNESCO World Heritage Book. The copper mines were then closed, but up to 1987 the smelting of copper in Sulitelma (at 67°08' of northern latitude), which had begun in 1886, was continued to be the greatest source of the emission of sulfur dioxide in Norway. At present, it is the production of nickel (organized in 1930's) in the vicinity of towns Nikel' and Zapolyarnyi (at the territory of Russia adjacent to its border with Norway, see Fig. 1) represents the largest source of the emission of SO₂ with strong negative effects on both sides of the border [11–16].

Earlier, corrosion tests were carried out in Sulitelma in the period from 1984 to 1988 [17]. Now, in the framework of the program of corrosion tests in the border area between Russia and Norway, the Norwegian Institute of Air Research (NILU) and the Institute of Physical Chemistry, Russian Academy of Sciences (IPC RAS), carried out two stages of more thorough investigations, namely, in 1990–1991 [18, 19], and in 1992–1994 [20, 21]. The main results of these tests are summarized in the present, conclusive report. For the first time, we are going to present the maps of the corrosion rate of plain carbon steel and zinc in the border

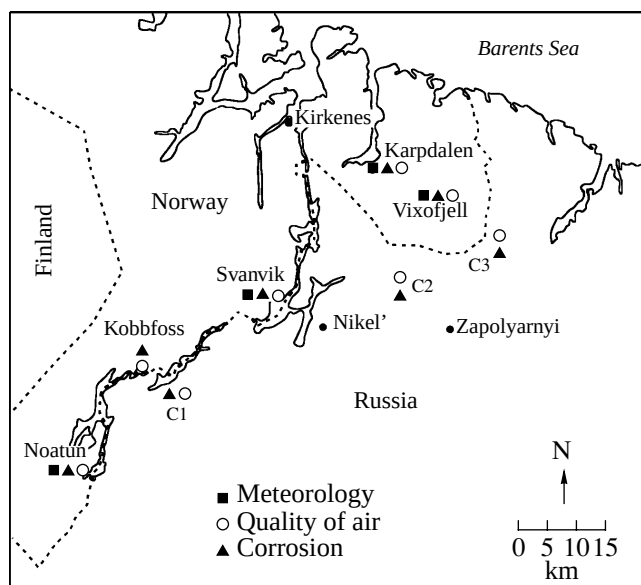


Fig. 1. Map showing the locations of test places and the types of the measurements performed there.

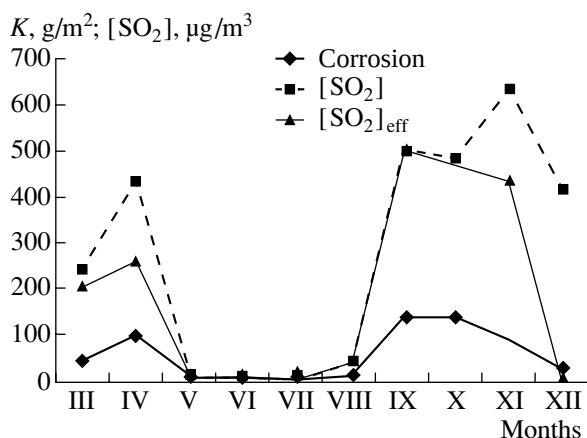


Fig. 2. The monthly corrosion of the carbon steel SO_2 , and the "effective" concentration of SO_2 for the temperature limit of corrosion -4°C .

areas of Norway and Russia (in terms of dose–response functions, derived concurrently with fulfilling the program, and the CONDEP dispersion model of emission [22, 23]), the maps of critical corrosion rates with two corrosion scenarios, and the of assessment of the tolerable (acceptable) levels of sulfur dioxide.

Table 1. Yearly corrosion rate ($\text{g}/(\text{m}^2 \text{ year})$) for three test places in Sulitelma

Period	Sandnes	Furulund	Lomi
January 1984–January 1985	592	493	—
January 1985–January 1986	449	431	—
March 1986–March 1987	549	511	488
March 1987–March 1988	67	60	66

TESTS IN SULITELMA

The tests were carried out at three places: Sandnes, Furulund and Lomi, concurrently with the Norwegian program for the observation of the environmental parameters. The Observation Program was closed in April 1987, and so was the production of copper with the smelting plants being exploded. At all the three places, the climatic parameters were virtually the same, whereas the monthly levels of SO_2 emission changed from the average values of $600 \mu\text{g}/\text{m}^3$ during winter months to $6 \mu\text{g}/\text{m}^3$ during the summer season, when the copper-smelting plants did not work. After stopping the production of copper, the SO_2 level in the region became virtually constant, about $2 \mu\text{g}/\text{m}^3$.

The corrosion part of the program included monthly tests of carbon steel, along with annual tests of steel, zinc, and copper. Comparing (see Table 1) the annual corrosion losses of steel in 1984 to 1986 and in 1987 (before and after closing the smelting plants, respectively), one clearly sees the dominant effect of SO_2 on the corrosion rate (which was then cut once and for all from nearly 500 to nearly $60 \text{ g}/(\text{m}^2 \text{ year})$). The corrosion rate of zinc was similarly decreased to $3 \text{ g}/(\text{m}^2 \text{ year})$. The obtained values agree well with other data for these metals at similar air temperature (Table 2). In the case of copper, the effect was less pronounced, since for the first year of exposure to unpolluted atmospheres copper oxidizes especially quickly due to the high content of ozone characteristic just of such areas [24, 25]. Only longer exposures can reveal the effect of SO_2 on copper in such a case.

For steel and zinc, we obtained simple regression equations relating their yearly corrosion (g/m^2) to the concentration of SO_2 ($\mu\text{g}/\text{m}^3$) in air:

$$ML_{\text{steel}} = 1.2[\text{SO}_2] + 63, \quad (1)$$

$$ML_{\text{zinc}} = 0.12[\text{SO}_2] + 4.1. \quad (2)$$

Needless to say that both the equations hold only for the given region, since climatic parameters (the same at all the above test sites) are implicitly included in the coefficients and constant terms of the equations. The climatic effect was investigated on the basis of the monthly tests of steel in 1986. Comparing the monthly mean concentrations of SO_2 with the corresponding monthly corrosion rates revealed the decrease in the effect of SO_2 with temperature, especially, during November–December of 1986 (see Fig. 2). The lowest corrosion rate was noted in December with its extremely low mean temperature (-10.2°C).

In order to take into account the temperature effect, a new parameter, "effective" concentration SO_2 (C_{eff}), was introduced. The daily concentration SO_2 (C_c) was reduced to C_{eff} via its multiplication by the fraction of the daily time t (h), during which the moisture film on

Table 2. Yearly corrosion of the carbon steel and zinc in cold and temperate climatic regions

Country, continent	Testing place	Climate	T, °C	Rh, %	Precipitations, mm	Steel, g/m ²	Zinc, g/m ²
Antarctica	Vanda	Very cold	−20	—	0.5–7	6.8	—
	Robertskollen	"	—	—	—	5.5	—
Russia	Oimyakon						
	1978/1979	"	−15...−17	66...72	123	8.1	1.8
	1987/1988	"	−15...−17	66...72	—	6.3	2.9
	1989/1990	"	−15...−17	66...72	—	4.7	1.5
	1990/1991	"	−15...−17	66...72	—	6.3	1.3
	Atka	"	−12	72	489	15	1.7
	Bilibino	"	−11	73	455	5.4	1.6
Canada	Norman Wells	Cold	−6		325	12	—
Russia	Klyuchi	"	−1	76	884	23	2.0
Sweden	Abisko	"	−1/−4	84	—	40	3.4
Norway	Sandnes	"	2.5	74	1056	67	3.0
	Furulund	"	2.5	74	1056	60	3.0
	Lomi	"	2.5	74	1056	66	4.0
Russia	Pobedino	"	−1	77	694	37	4.3
Norway	Noatun	"	0	77	—	78	5.4
Sweden	Gallivare	"	1		527	47	5.7
Russia	Yakovlevka	"	2	70	410	41	4.6
	Pogranichnyi	Temperate	4	67	417	49	4.3
	Zvenigorod	"	5	82	600	110–140	6.6
Sweden	Aspvreten	"	6	83	543	81–147	6.0–8.2
Germany	Garmish-Partenkirchen	"	8	82	1492	86–133	3.7–8.3

the surface is unfrozen:

$$C_{\text{eff}} = C_c(t/24).$$

For the purpose of determining the threshold, a regression analysis of the corrosion losses as a function of the “effective” concentration of SO₂ was carried out for the supposed temperature thresholds of 0, −2, −4, and −6°C (Table 3). According to the results obtained, the sought melting threshold lies between −2 and −4°C.

TESTS IN THE AREAS ALONG THE NORWAY–RUSSIA BORDER

The test area along the Norway–Russia border (Fig. 1) is considerably larger than the environs of Sulitelma. The climatic conditions vary inside the area, and the effect of marine aerosols is pronounced in its northern part [18–21].

The annual corrosion rates of steel and zinc for a period of 1990–1991 are listed in Table 4. At all the test sites along the border, they were higher than in Sulitelma upon ceasing the copper smelting (i.e., 60 and 3 g/(m² year) for steel and zinc, respectively). In the maximum, that is, in Vixofjell and at the site C2, the

rates were 5.8 and 5.1 times that in Sulitelma for steel and 8 and 7 times that for zinc. The average annual levels of SO₂ at these places were 37.1 and 56.9 µg/m³, respectively. Figure 3 shows the annual corrosion rates of the carbon steel in the period of 1990–1991 versus the concentration of SO₂ for the test sites in the Norway and Russia. An additional effect of chlorides on the corrosion rate in Vixofjell (Norway) is easily seen. The

Table 3. Linear regression equations for monthly corrosion in Sulitelma as a function of the “effective” concentration of SO₂, corresponding to different freezing points of the surface moisture and controlling the effect of SO₂

Freezing point, °C	Without accounting for the freezing point	R ²
Linear regression equations	$ML = 0.19 \text{ SO}_2 + 9.1$	0.74
0	$ML = 0.27 \text{ SO}_2 + 20.8$	0.84
−2	$ML = 0.27 \text{ SO}_2 + 13.2$	0.94
−4	$ML = 0.24 \text{ SO}_2 + 13.1$	0.92
−6	$ML = 0.27 \text{ SO}_2 + 13.1$	0.89

Note: *ML* is the weight loss, g/m².

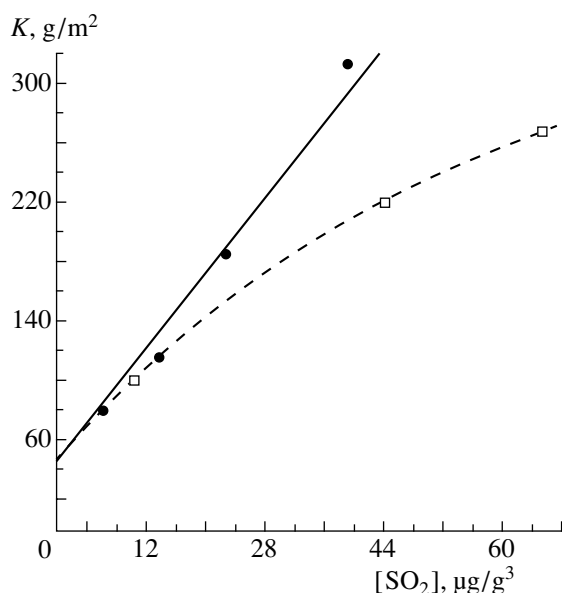


Fig. 3. The yearly weight loss of the carbon steel as a function of the mean annual concentration of sulfur dioxide: (●) the places on the territory of Norway and (□) the places on the territory of Russia.

presence of chlorides was confirmed by the analyses of filters that were each month exposed with the aid of a special collector of aerosols, developed at the NILU. The quantity of chlorides in Vixofjell was found to be four times the mean value of them in Svanvik.

Two-year tests (August 1992–July 1994) at two sites in Norway and three sites in Russia (Table 5) showed that corrosion both of steel and zinc decelerate with time; remaining yet high at the sites with high SO_2 level (Vixofjell, C2 and C3).

The monthly tests confirmed that, like in Sulitelma, the atmospheric corrosion occurs along the Norway–Russia border even in winter, when the temperature is subzero during the whole month. The results of the monthly tests of carbon steel were processed similarly to those obtained in Sulitelma. In this case, however, the temperature effect was estimated by selecting various temperature thresholds for the time of wetness (TOW). From Table 6, it is seen that the value of R_2 is maximum when counting TOW from the temperature threshold -4°C . Thus, both the series of tests show

approximately the same freezing point of electrolyte on the metal surface. It should be noted that equations in Table 6 have a negative constant term, and, therefore, can be used only in the range of the environmental parameters in which they have been obtained.

On completing the second phase of atmospheric tests of materials, a similar approach was applied to processing the dose-response functions, but on the basis of annual tests. The test sites were subdivided into two groups. Group A comprised all sites in Norway and the nearest to the border site C1 in Russia (see Fig. 1). All these test areas were affected by marine aerosols (see the upper curve in Fig. 3). Group B comprised all the test sites Norway and Russia slightly polluted with chlorides (the lower curve in Fig. 3). The data of the steel and zinc exposures in 1990–1991 and 1992–1993 were analyzed. The time of wetness was measured from the temperature thresholds 0, -2 , and -4°C .

This subdivision of the test sites into two groups enabled us to obtain, by the regression analysis, the dependences of the corrosion rates of steel and zinc on the environmental parameters with the predicting probability R^2 in a range from 0.974 to 0.994 for the models

$$ML = A + B \cdot TOW[\text{SO}_2] \quad (3)$$

and

$$ML = (A + B \cdot [\text{SO}_2])TOW, \quad (4)$$

respectively.

As was already noted in [26], such a high probability of (prognosis) is characteristic of simulation in mesoregions with small climatic variations. In macroregions, significant climatic differences and the presence of other corrosive pollutants in the atmosphere reduce, as a rule, the probability of corrosion prediction.

Based on the data of B group, we derived equations for those (southern and the south-western) areas, where SO_2 is the dominant pollutant, while the data of A group resulted in equations for the (northern and the north-eastern) areas partially affected by chlorides. Directly taking chlorides into account in the dose-response functions would certainly be useful. However, regular measurements of dry chloride precipitations, could be performed only in Vixofjell, Karpdalen, and Svanvik.

To simulate corrosion situation in the region, we used the equations in the form $ML = A + B \cdot TOW[\text{SO}_2]$,

Table 4. Yearly corrosion rates of carbon steel and zinc, $\text{g}/(\text{m}^2 \text{ year})$ at the test places along the Russian–Norwegian border in the periods of 1990–1991 and 1991–1992

Period, years	Material	Vixofjell	Karpdalen	Svanvik	Noatun	S1	S2	S3
1990–1991	Steel	308	180	108	78	93	261	214
1992–1993	Steel	347	—	145	—	163	306	264
1990–1991	Zinc	24	12	9.6	5.4	6.4	19	15
1992–1993	Zinc	18	—	7.1	—	8.3	21	15

Table 5. Weight losses of the carbon steel and zinc, g/m², at the test places along the Russian–Norwegian border after one (August 19, 1992–July 19, 1993) and two (August 19, 1992–July 19, 1994) test years

Material	Vixofjell		Svanvik		S1		S2		S3	
	1	2	1	2	1	2	1	2	1	2
Steel	347	539	145	206	163	220	306	466	264	392
Zinc	18.0	26.8	7.1	9.5	8.3	11.2	21.3	33.4	15.1	27.1

taking into account that corrosion starts from the temperature of -4°C .

JOINT ASSESSMENT OF THE RESULTS

Using the approach [23], as well as the data obtained in Sulitelma and along the Russia–Norway border, as the basis, we developed a model estimating the monthly corrosion of carbon steel. The model is based on the following assumptions.

(1) Dry precipitations make the main contribution to the atmospheric corrosion of metals, especially in cold climatic regions, where the main mass of precipitations consists of dry snow. The equation describing dry SO_2 precipitation, i.e.,

$$f_{\text{dry}} = c[\text{SO}_2]^A TOW^B, \quad (5)$$

is based on a correlation between the corrosion of metal under the adsorption films of water and the quantity of SO_2 , absorbed by the film for a time equal to TOW .

(2) TOW itself was replaced by the average monthly temperature (T) and the relative humidity of air (Rh). The high correlation between these factors, as well as the possibility of quantitatively describing TOW through the monthly or annual mean values of Rh and T , was shown in [27, 28]; the dependence of TOW (and, accordingly, corrosion) on Rh may be exponential or close to linear, whereas the effect of T is more complex. Both TOW and corrosion increase with T . At certain (not necessarily the same) values of T , TOW and corrosion rate reach their maxima. With a further increase in temperature TOW descends (see Fig. 4b) because of diminishing Rh (the negative correlation between Rh and T). Moreover, corrosion can also be affected by accelerated evaporation of the water films because of the radiation heating of metal.

(3) In describing the effect of temperature (considering the cold climate), we took into account the threshold of corrosion effects lying below 0°C . As a rule, this slightly affects the yearly sum of TOW , but monthly sums (according to the data obtained a long the Russia–Norway border) can change by 500 times. Figure 4a shows the temperature dependences of TOW for the thresholds 0 and -4°C . In the low range of temperatures, the TOW curve for $T > -4^{\circ}\text{C}$ shifts to the left, while starting from approximately 0°C , one may suppose the existence of a temperature range, in which corrosion is independent of T .

In view of the aforesaid, we used a nonlinear regression analysis to derive dose–response functions. For this purpose, we created a body of 111 sets of monthly data. The monthly corrosion of steel was described by a system of three equations:

$$ML = 0.084[\text{SO}_2]^{0.53} \exp\{0.053Rh + 0.1(T - 1)\}, \quad (6)$$

for $T \leq 1^{\circ}\text{C}$,

$$ML = 0.084[\text{SO}_2]^{0.53} \exp\{0.053Rh\}, \quad (7)$$

for $1 < T \leq 5^{\circ}\text{C}$,

$$ML = 0.084[\text{SO}_2]^{0.53} \exp\{0.053Rh - 0.036(T - 5)\}, \quad (8)$$

for $T > 5^{\circ}\text{C}$,

where ML is the weight loss in g/m²; SO_2 is the concentration in $\mu\text{g}/\text{m}^3$; Rh is the relative humidity, in %; and T is the temperature of air.

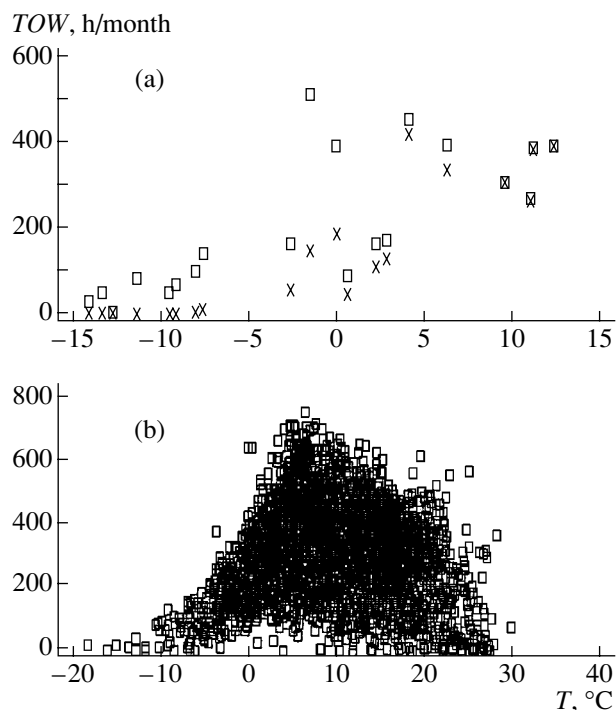


Fig. 4. The monthly time of wetness versus temperature: (a) the data obtained in the areas of Sulitelma and along the Russian–Norwegian border (X, TOW has been determined for $T > 0^{\circ}\text{C}$; $\square$, TOW , for $T > 4^{\circ}\text{C}$); (b) the data of the UN/ECE program (TOW has been determined for $T > 0^{\circ}\text{C}$).

Table 6. Coefficients of the regression equation $ML = (A + B \cdot SO_2 + C \cdot Cl) \cdot TOW$ and the value of R^2 for the time of wetness with various temperature limits

Temperature threshold TOW , °C	A	B	C	R^2
> -2	-0.26	0.0088	0.0001	0.79
> -4	-0.16	0.0077	0.00004	0.96
> -6	-0.07	0.0067	0.00001	0.81

Note: ML is the weight loss, g/m^2 .

Thus, the monthly corrosion of steel increases with the monthly temperature up to 1°C, does not change in a temperature range from 1 to 5°C, and exhibits some decrease with its further growth. The maximum is reached already at 1°C instead of 5°C following from traditional calculation (i.e., by 4°C lower). This fact well correlates with the maximum value of TOW calculated for $T > -4^\circ\text{C}$ and $Rh > 80\%$.

CORROSIVITY OF THE ATMOSPHERE ALONG THE NORWAY–RUSSIA BORDER

Recently, great attention has been focused on the mapping of territories with respect to the levels of the corrosive pollution of air, including such mesoregions, which involve a number of SO_2 emission sources. Corresponding calculation techniques and programs have been developed, taking into account climatic parameters [22]. This turned to be very useful in mapping the negative effects of the pollution of atmosphere, in particular, with sulfur dioxide, on various receptors of the environment (e.g., aqueous ecosystems, forests and natural vegetation, agricultural plants, and human health). This group also includes the effects of air pollution on the materials, since the corrosivity of atmosphere as an important factor of the corrosion risk predetermines in many respects the strategy of reducing the level of pollution.

The concentrations of SO_2 were simulated in the 100×100 km region along the border. The CONDEP model for 5×5 km grid squares was employed to calculate the concentrations of SO_2 . The model uses a long-term frequency matrix of wind directions and classes of its stability, and allows taking into account 50 of point emission sources. In calculating the areas along the border, 11 strong and 31 weak emission sources were taken into account, mainly at the Russian territory.

For the southern area of the region, Eqs. (9) and (10) from B group were used:

$$ML_{\text{steel}} = 69.1 + 12.2TOW[SO_2], \quad R^2 = 0.974, \quad (9)$$

$$ML_{\text{zinc}} = 4.40 + 0.90TOW[SO_2], \quad (10)$$

$$R^2 = 0.976,$$

while for the northern part where the effects of marine aerosols are more perceptible, Eqs. (11) and (12) from A group:

$$ML_{\text{steel}} = 27.5 + 26.5TOW[SO_2], \quad R^2 = 0.989, \quad (11)$$

$$ML_{\text{zinc}} = 2.40 + 1.60TOW[SO_2], \quad (12)$$

$$R^2 = 0.879.$$

Here, ML is the yearly loss of weight in g/m^2 ; TOW is the portion of the yearly time, combining temperature $T > -4^\circ\text{C}$ with a relative humidity $Rh > 80\%$, while SO_2 is the average yearly concentration of sulfur dioxide, in $\mu\text{g}/\text{m}^3$.

Figures 5 and 6 show the mapping results for steel and zinc, respectively. The concentrations of SO_2 are taken from the results obtained in the 1998 summer and 1998–1999 winter seasons [21].

The maps for both the materials are similar to those with the highest corrosion rates in Nikel' and Zapolarnyi areas, where the integrated nickel-production plants are located. The highest SO_2 emission and yearly corrosion rates are observed just in Nikel': 457 and 33 g/m^2 for steel and zinc, respectively. The lines of equal corrosion rates are plotted with intervals of 30 $g/(m^2 \text{ year})$ for steel and 3 $g/(m^2 \text{ year})$ for zinc. From these graphs, it is seen that the prevailing north-eastern and northern directions of wind predetermine the elevated corrosivity of atmosphere in this directions at considerable distances from the SO_2 emission sources, including those in Pechenga, Vixofjell and Karpdalen.

The equations used include only SO_2 and TOW as corrosion-determining factors. That is why the calculated corrosion rates are valid for the areas that are not affected by a substantial amounts of marine aerosols. In the coastal areas and in the sea, the above atmospheric corrosion rates should be considered as a partial contribution of SO_2 to the general corrosion rate.

The prepared maps seem to be useful in estimating the corrosivity of atmosphere in the region, and its trend with time.

ACCEPTABLE CORROSION RATES AND SULFUR DIOXIDE LEVELS

The atmospheric corrosion of metals is a cumulative, irreversible process, which can also occur in the absence of impurities. Yet even a small pollution of air, in particular with sulfur dioxide, makes corrosion increase. That is why the approach applied to ecosystems is unsuitable for materials, because it uses critical levels/loads, on exceeding of which irreversible changes in receptors take place. Instead of it, we developed a conception of tolerable (or acceptable) corrosion rates and pollution levels. The conception was found useful for estimating the effects of acidic pollution of air on the materials within the framework of the

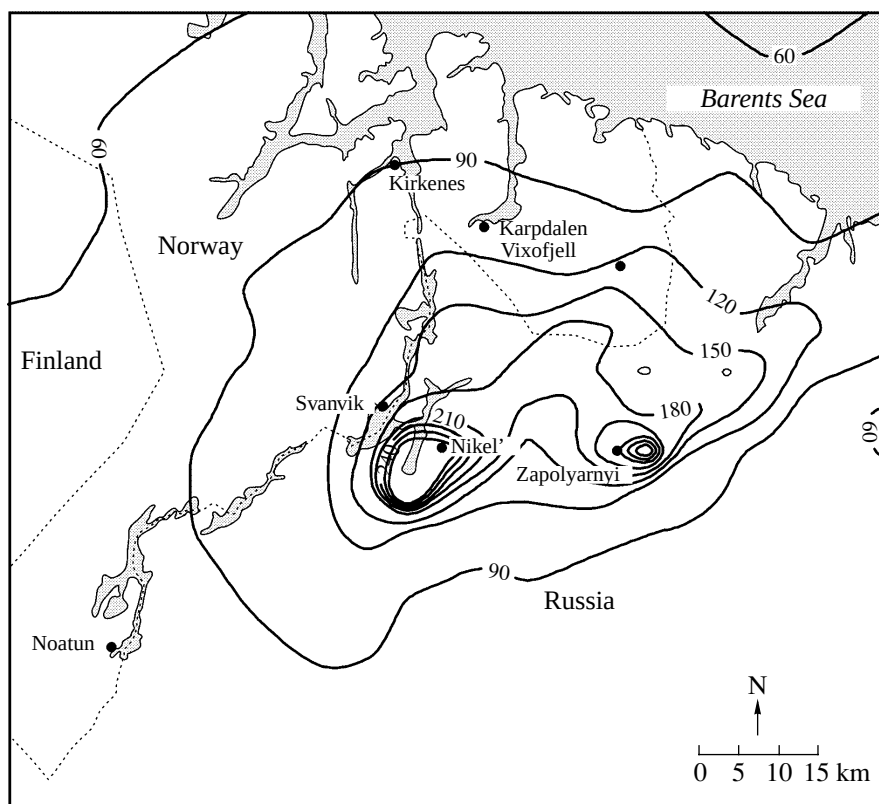


Fig. 5. The map of the yearly corrosion of the carbon steel in the border areas of Russia and Norway. The simulated region measures 110×100 km.

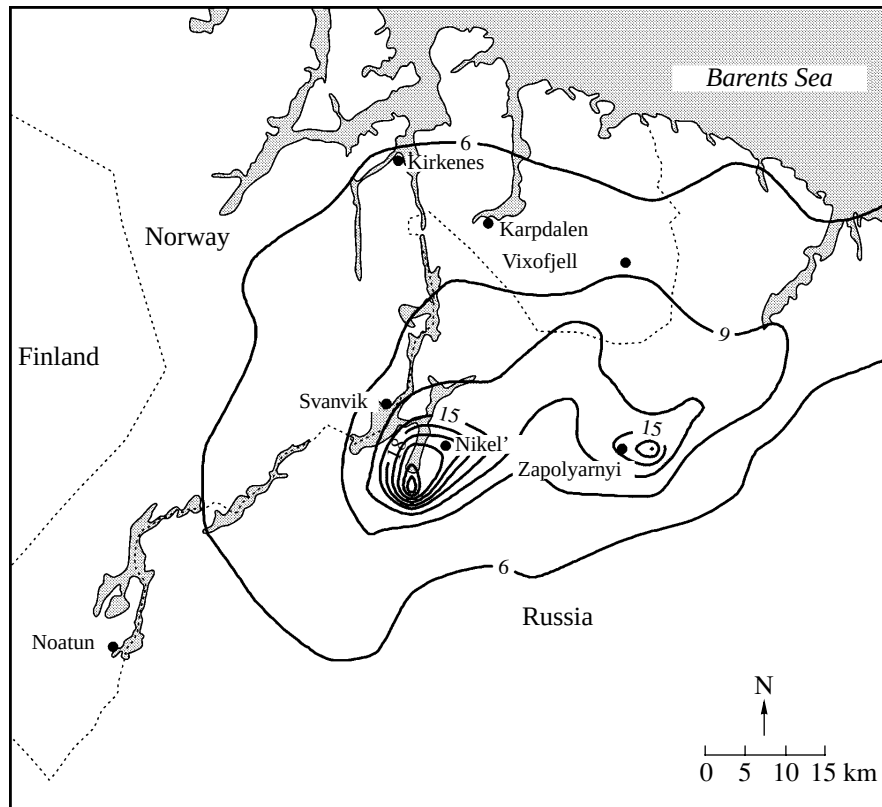


Fig. 6. The map of the yearly corrosion of zinc in the border areas of Russia and Norway. The simulated region measures 110×100 km.

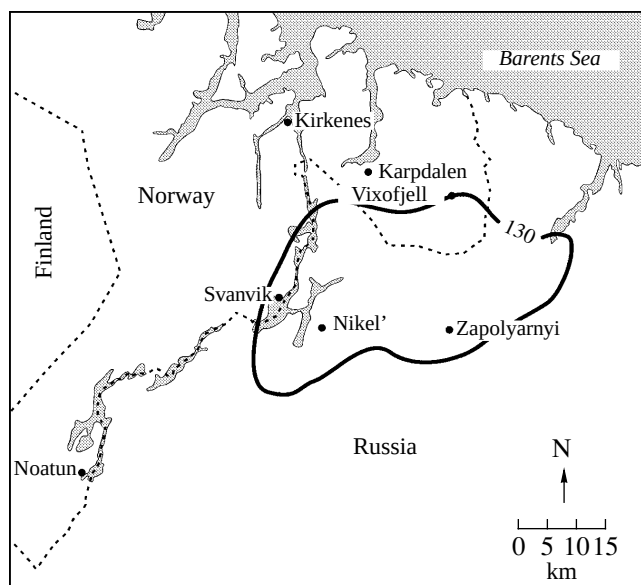


Fig. 7. The atmospheric corrosivity in the border areas of Russia and Norway with respect to the carbon steel for the scenario $n = 2$; the contour on the map shows the acceptable corrosion rate of steel ($130 \text{ g/(m}^2 \text{ year)}$) which is twice as high as the background one; the territory inside the contour has the corrosion rate higher than the acceptable one; and the territory outside the contour, lower than the acceptable one.

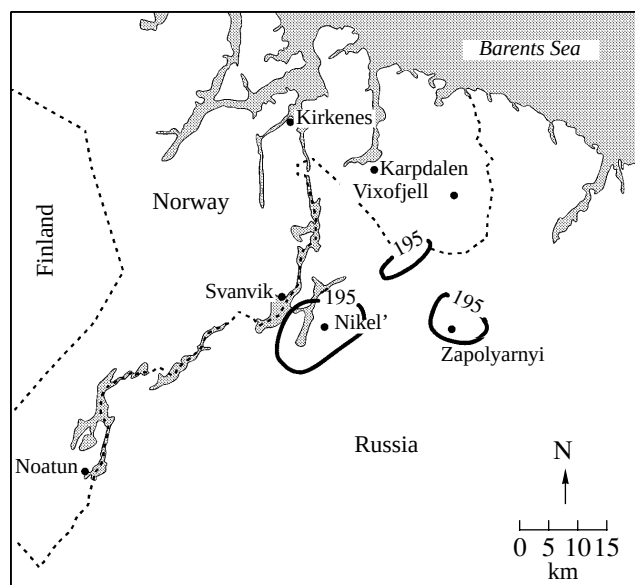


Fig. 8. The atmospheric corrosivity in the border areas of Russia and Norway with respect to the carbon steel for the scenario $n = 3$; the contour on the map shows the acceptable corrosion rate of steel ($195 \text{ g/(m}^2 \text{ year)}$), which is three times higher than the background one; the territory inside the contours has the corrosion rate higher than the acceptable one; and the territory outside the contours, lower than the acceptable one.

Convention on the transboundary pollution of air at long distances (as accepted by UN/ECE). It comprises the mapping of the areas with an enhanced corrosion risk, and features different scenarios of the pollution of the atmosphere [24, 29, 30].

The acceptable corrosion rate is a quantity selected on the basis of a technical or economical (expert) examination with a view of specific application of materials. However, for model calculations and comparison of the sensitivity of various materials to the air pollution, the tolerable (acceptable) levels can be determined through their ratio to the corrosion rate in the atmosphere with the “background” pollution:

$$n = K_a/K_b = K_a/K_{10\%}, \quad (13)$$

where K_a is the tolerable (acceptable) corrosion rate; and K_b is the “background” corrosion rate, calculated as 10 percentile from the body of the data of the UN/ECE program ($K_{10\%}$). It should be noted that the “background” corrosion rates, determined for the 1997–1998 yearly period, are selected constants, characteristic of a given metal (65 and to $3.3 \text{ g/(m}^2 \text{ year)}$ for steel and zinc, respectively). This is very important, since, in fact, the rates in background regions substantially depend on the climatic parameters, in particular, temperature and relative humidity of air (see Table 2) [24, 30].

Given the dose–response function and the fixed value of n , the tolerable level of sulfur dioxide can be determined as

$$nK_{10\%} = f([\text{SO}_2]_{\text{acc}}[TOW]). \quad (14)$$

In particular, for Eq. (3) we have in a general form:

$$[\text{SO}_2]_a = (nK_{10\%} - A)/(B[TOW]). \quad (15)$$

Hence, the critical level of sulfur dioxide depends on the time of wetness, that is (when using the UN/ECE function [24, 29]), on T and Rh . Since $K_b = K_{10\%}$ is constant for a metal, the acceptable levels of $[\text{SO}_2]$ will be lower for the regions with higher $[TOW]$ and with the UN above ECE functions, lower for the temperate climate compared to colder or warmer ones.

Figures 7 and 8 show the maps of the atmospheric corrosivity for steel along the Norway–Russia border at tolerability factors n equal to 2 and 3, and, accordingly, at corrosion rates of 130 and $195 \text{ g/(m}^2 \text{ year)}$. It is easily seen how considerably differ from each other areas, at which the corrosion rate exceeds the indicated limits (the areas inside the contours). The area $n = 2$ ($>130 \text{ g/(m}^2 \text{ year)}$) comprises also a part of the Norway territory. The area with the corrosion rate of more than three times the background one (Fig. 8) is fragmentary, its northern fragment, in the vicinity of the border line, having no SO_2 emission source. The pollution of its atmosphere is most probably due to the relief of the terrain (in particular, to the location of this area lower than

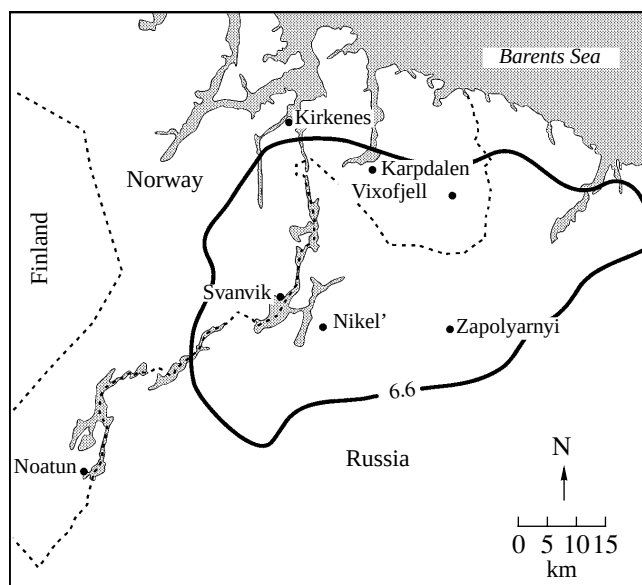


Fig. 9. The atmospheric corrosivity in the border area of Russia and Norway with respect to zinc for the scenario $n = 2$: the contour of the map shows the acceptable corrosion rate of zinc ($6.6 \text{ g}/(\text{m}^2 \text{ year})$), which is twice as high as the background one; the territory inside the contour has the corrosion rate which is higher than the acceptable one; and the territory outside the contour, lower than the acceptable one.

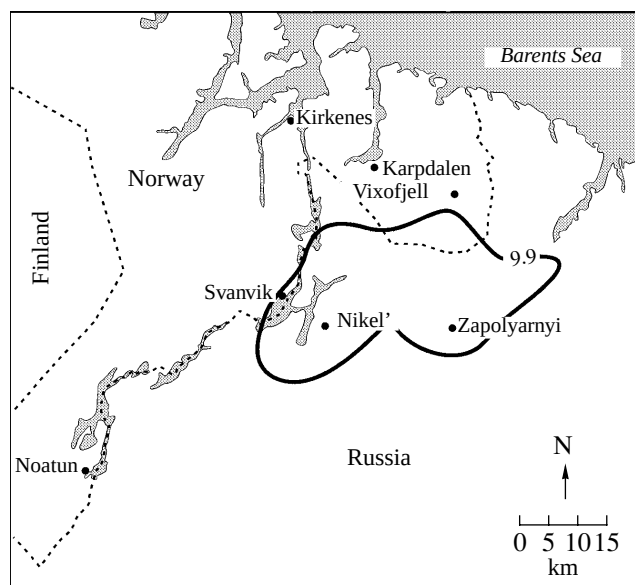


Fig. 10. The atmospheric corrosivity of the border areas of Russia and Norway with respect to zinc for the scenario $n = 3$: the contour shows the acceptable corrosion rate of zinc ($9.9 \text{ g}/(\text{m}^2 \text{ year})$) which is three times higher than the background one; the territory inside the contour has the corrosion rate higher than the acceptable one; and the territory outside the contour, lower than the acceptable one.

the SO_2 emission sources in Nikel' and Zapolyarnyi) and to the predominant wind direction. The tolerable coefficient $n = 2$ (or 3) is attained with a tolerable SO_2 level of 12.9 (or 21.0) $\mu\text{g}/\text{m}^3$ in the northern of the region, and 16.6 or 34.4 $\mu\text{g}/\text{m}^3$ in the southern area, respectively. Evidently, if due to a decrease in the emission of sulfur dioxide, its level in the curves in Fig. 8 diminishes in the northern area from 21.0 to 12.9 $\mu\text{g}/\text{m}^3$, the corrosion rate on these curves will drop to a level corresponding to $n = 2$ ($130 \text{ g}/(\text{m}^2 \text{ year})$), the area where K_a exceeds that allowable for $n = 2$, will considerably reduce. In the southern area of the region (Fig. 8), a similar effect will be reached with a decrease in the SO_2 emission from 34.4 to 16.6 $\mu\text{g}/\text{m}^2$.

Both in Russia and Norway, the areas in which the acceptable corrosion rates of zinc are exceeded for $n = 2$ and 3 (Figs. 9 and 10) are noticeably larger than those for steel. The acceptable SO_2 emission levels were 8.75 and 15.6 $\mu\text{g}/\text{m}^3$ in the northern area, but 12.7 and 20.4 $\mu\text{g}/\text{m}^3$ in the southern areas, for $n = 2$ and 3, respectively. In other words, for zinc, the acceptable SO_2 levels are lower, while their intervals in which the acceptable corrosion rates increase from $n = 2$ to 3 are narrower in both the areas in question. The main reason for this is presumably that the real background and acceptable (at $n = 1$) corrosion parameters do not give coinciding isokorrods, and the calculated $[\text{SO}_2]_a$ levels do not coincide either. By and large, the shown maps well illustrate the approach applied to determine the

acceptable corrosion rates and SO_2 emission levels. This approach seems to be useful in assessing corrosion situations and developing scenarios of counteracting the emission of air pollutants.

CONCLUSIONS

The investigations carried out in Sulitelma and along the Russia–Norway border with the use of the “effective SO_2 concentration” and negative temperature thresholds of TOW , respectively, led us to virtually the same conclusion: the temperature threshold of the corrosion development lies between -2 and -4°C .

The annual corrosion data for all the tested materials well correlate with the SO_2 emission levels in Sulitelma, and with SO_2 and TOW in the areas along the Russia–Norway border. All the equations derived can be used for the simulation of the corrosivity on these territories.

In terms of the approach suggested in the UN/ECE program on materials, we developed a model of monthly corrosion involving the monthly-mean temperature and relative humidity of air instead of the time of wetness. The corrosion is described by three Eqs. (6)–(8) for three ranges of the average monthly temperature. The threshold $T_{\text{month}} \leq 1^\circ\text{C}$ is comparable to the freezing point of water films, detected in both the investigations carried out in the areas where the effect of SO_2 dominates.

The existing equations describe the corrosion of metals as a function of the parameters of the surrounding medium. The equations for the first year of the corrosion of carbon steel and zinc in the areas along the Russia–Norway border hold, given the concentration of SO_2 and TOW . The simulation of the distribution of SO_2 became practicable on the basis of dispersion models. The industrial CONDEP module was used for this purpose in the areas along the border. Other models may be used for other regions and big cities.

The corrosion rates (including the acceptable ones) for carbon steel and zinc are mapped for the region along the Russia–Norway border of the total area of 110×100 km to the scale of 5×5 km for two corrosion scenarios. The corresponding tolerable (acceptable) SO_2 levels have been determined.

All the investigations revealing the environmental effects of sulfur dioxide in the areas along the Russia–Norway border resulted in June of 2001 in a practically important decision on the reconstruction of the integrated nickel-production works.¹

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¹ The Norwegian Government provided a financial support to Russia for the reconstruction of the integrated nickel-production works by granting \$30 million.

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